

Candidate resolutions for interval and absolute-value problems

Seven proposed resolutions in OpenProblemsInNLA

This reader contains six self-contained manuscripts addressing seven entries in the repository section on intervals and absolute value equations. The arguments are presented as claimed resolutions for independent review. They have not been accepted by the repository, independently peer reviewed, or externally submitted by this work.

Claim index

- AV-01** Exactly 2^n solutions of $Ax + |x| = b$ can be recognized with $n+1$ rational LP feasibility tests, without a regularity promise.
- AV-02** The spectral inverse-norm threshold is NP-hard under a many-one reduction, even for integer upper-triangular matrices with diagonal 2. The decision problem on the corresponding rational class is NP-complete.
- IV-02** Exact tridiagonal interval determinant ranges are NP-hard to compute, even for regular interval families. The upper determinant threshold is NP-complete.
- IV-03** An interval matrix is inverse-M throughout if and only if n^2 specified vertices are inverse-M. This implies the proposed $2n^2$ criterion.
- IV-04** Exact tridiagonal interval solution hulls are NP-hard to compute, even with a regular interval family and point right-hand side $-e_n$.
- IV-05** An exact solution hull under the inverse-M interval promise is computable with $2n$ rational linear programs.
- IV-06** A 3×3 interval matrix with integer endpoints has at least four connected components in its real eigenvalue set.

AV-03 and IV-01 are not claimed solved. The tridiagonal hardness result is a bit-complexity result and is not claimed to be strong NP-hardness or an approximation lower bound. The general exact-output questions for IV-02 and IV-04 are equivalent to $P = NP$, as explained in their joint manuscript.

Reviewing the package

Start with IV-06 for a short, exact counterexample. Its certificate consists of four integer eigenpairs and three excluded real numbers. AV-01 gives the shortest new structural characterization. One rational-rotation reduction then supplies both IV-02 and IV-04. The inverse-M recognition and hull arguments in IV-03 and IV-05 are independent except that the former can check the promise needed by the latter.

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